VERY EASY & CUTE SECTION WITH MANY MINI ANSWERS

VSAQ

SECTION-B

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VSAQ CHAPTERS

28 MARKS

★★★ 1.	SOLUTIONS	(2+2 Marks)	56 - 57
★★★ 2.	ELECTRO CHEMISTRY	(2+2 Marks)	58 - 59
★★ 3.	CHEMICAL KINETICS	(2 Marks)	60 - 61
★★★ 4.	THE d&f -BLOCK ELEMENTS	(2+2 Marks)	62 - 63
★★ 5.	COORDINATION COMPOUNDS	(2 Marks)	64 - 65
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★★★ 7.	ALCOHOLS, PHENOLS &ETHERS	(2 Marks)	68 - 69
★★ 8.	ALDEHYDES, KETONES	(2 Marks)	70 - 71
★★ 9.	AMINES	(2 Marks)	72 - 73
★★★ 10.	BIOMOLECULES	(2+2 Marks)	74 - 75

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Super Scoring Pages

★★★SSP★★★

Page 56,57,58,59,60,62,63,64,66,68,70,72,74,75

Total SS Pages : 14

1 SOLUTIONS

2+2Marks

75. Define molarity.

A: **1) Molarity (M):** It is the 'number of moles of the solute' dissolved 'per litre' of the solution.

2) Formula: $\text{Molarity(M)} = \frac{\text{No.of moles of solute}}{\text{Volume of solution in Litre}}$

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76. Define molality. [AP 15]

A: **1) Molality(m):** It is the 'number of moles of the solute' dissolved in 'one kilogram (Kg) of the solvent'

2) Formula: $\text{Molality(m)} = \frac{\text{No.of moles of solute}}{\text{Mass of solvent in Kg}}$

---

77. Define mole fraction. [AP 16,18,19]

A: **1) Mole Fraction(χ):** It is the 'ratio of the number of moles of one component' to the total number of moles of all the components present in the solution.

2) Formula: Mole fraction $\chi = \frac{\text{No.of moles of one component}}{\text{Total no.of moles of all the components}}$

---

78. State Henry's law. [BMP][AP 18,20]

A: **1) Henry's law:** The partial pressure (p) of a gas in vapour phase is directly proportional to the mole fraction (χ) of the gas in the solution.

2) Formula: Partial pressure of a gas $p_{gas} = K_H \chi_{gas}$.

☺ Henry said $p = K_H \cdot \chi$

---

79. Define osmotic pressure. [BMP][AP 15,16,18]

A: **1) Osmotic pressure (Π)** is the pressure required just to stop the Osmosis. (or)
It is the minimum pressure required to stop the flow of solvent particles through a semipermeable membrane from a pure solvent into a solution.

2) Formula: $\Pi = CRT$

---

80. What are isotonic solutions? [AP 17,22,24]

A: **1) Isotonic solutions** are the solutions which have the 'same osmotic pressure' at a given temperature .

2) Ex: Blood is isotonic with 0.9% (w/v) NaCl.

81. What is Ebullioscopic constant?

A: **1) Ebullioscopic constant(K_b) :** It is the **elevation in boiling point** of a solution when one mole of non-volatile solute is added to 1kg of solvent.

2) Formula: $\Delta T_b = K_b m$ **Units:** $k.kgmol^{-1}$

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82. What is Cryoscopic constant? [AP 23]

A: **1) Cryoscopic constant (K_f):** It is the **depression in the freezing point** of a solution when one mole of non-volatile solute is dissolved in 1kg of solvent.

2) Formula: $\Delta T_f = K_f \times m$ **Units:** $k.kgmol^{-1}$

---

83. Calculate the amount of benzoic acid (C_6H_5COOH) required for preparing 250 mL of 0.15 M solution of benzoic acid in methanol.

A: Molarity(M)= 0.15M, Volume (V)=250mL,

GMM (C_6H_5COOH) =122, Weight of C_6H_5COOH (w) =?

GMM of C_6H_5COOH
$(12\times6)+(5\times1)+(12\times1)+(16\times2)+(1\times1) = 122$

Formula: Molarity $M = \frac{w}{GMM} \times \frac{1000}{V(mL)}$

$$\Rightarrow w = \frac{M \times GMM \times V(mL)}{1000} = \frac{0.15 \times 122 \times 250}{1000} = 4.575 \text{ g}$$

---

84. Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution. [AP 15]

Sol: Given weight of solute (NaOH) is w=5 g; GMM of NaOH =40

Moles of NaOH, $n = \frac{weight}{GMM} = \frac{5}{40}$

Volume of the solution in mL = 450 mL, Molarity=?

☺ EASY PROBLEM ☺

Molarity $M = n \times \frac{1000}{V(mL)} = \frac{5}{40} \times \frac{1000}{450} = 0.278 \text{ mol dm}^{-3}$

---

85. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene.

Sol: Given weight of ethanoic acid = 2.5g

GMM of ethanoic acid $C_2H_4O_2 = (12\times2)+(1\times4)+(16\times2) = 60$

Moles of $C_2H_4O_2$, $n = \frac{weight}{GMM} = \frac{2.5}{60}$

weight of solvent (benzene)= 75 g Molality (m) =?

☺ IMP. PROBLEM ☺

$$\text{Molality(m)} = n \times \frac{1000}{\text{Mass of solvent (in g)}} = \frac{2.5}{60} \times \frac{1000}{75} = 0.556 \text{ mol.kg}^{-1}$$

2 ELECTRO CHEMISTRY

2+2Marks

86. State Faraday's first law of electrolysis. [BMP][AP 16,18,20,25]

A: Faraday's first law of electrolysis:

"The mass (m) of the substance deposited or liberated at the electrode is directly proportional to the quantity of electric charge (Q) passing through the electrolyte".

Thus, $m \propto Q$.

☺ FSL ☺ FIRST BEST 'Q'

---

87. State Faraday's second law of electrolysis. [AP 20]

A: Faraday's second law of electrolysis:

"When same quantity of electric charge is passed through different electrolytes, connected in series the amount of different substances deposited at the electrodes is directly proportional to their equivalent masses".

Mathematically, $\frac{m_1}{m_2} = \frac{E_1}{E_2}$

☺ FSL ☺ SECOND BEST 'Q'

---

88. What is a fuel cell? How is it different from a conventional galvanic cell?

A: 1) **A fuel cell** is a device that converts combustion energy of fuel directly into electrical energy.

2) Conventional galvanic cell converts chemical energy of the spontaneous redox reaction into electrical energy.

3) Fuel cells relatively generate less pollution when compared to conventional galvanic cells.

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89. What is metallic corrosion? Give one example. [BMP] [AP 15]

A: 1) **Metallic Corrosion:** It is the process of slow eating up of metals by gases and water vapours present in atmosphere due to the formation of certain compounds like oxides, sulphides, carbonates,etc

2) **Ex:** Rusting of Iron (Fe converts into its oxide)

☺ Even Corrosion gets U 2 Marks! ☺

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90. Give the electro-chemical reaction that represents the corrosion or rusting of iron.

A: 1) Corrosion reaction:

Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ **Anode:** $2Fe \rightarrow 2Fe^{+2} + 4e^-$

2) Overall reaction: $2Fe + O_2 + 4H^+ \rightarrow 2Fe^{+2} + 2H_2O$

91. Define molar conductivity Λ_m and how is it related to conductivity (k)?

A: 1) **Molar conductivity** is defined as the conductance of the solution containing one gram-mole of the electrolyte, when the electrodes are unit distance apart.

2) Molar conductivity $\Lambda_m = \frac{k \times 1000}{M}$. Here k is conductivity, 'M' is molarity.

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92. What is cell constant of a conductivity cell?

A: 1) **Cell constant (*l*/a)** is the ratio of distance between electrodes and area of cross section.

2) Its unit is Cm^{-1} or m^{-1}.

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93. How is a galvanic cell represented on paper as per IUPAC convention? Give one example.

A: 1) As per IUPAC convention, a galvanic cell is represented as follows.
For the Daniel Cell : $Zn/Zn^{+2}(C_1)||Cu^{+2}(C_2)/Cu$

2) The electrode on which oxidation takes place is written on the left hand side and the other electrode on which reduction takes place is written on the right hand side.

3) Oxidation electrode and reduction electrode are separated by double line which represents salt bridge. C_1 and C_2 represent concentrations of Zn^{+2} and Cu^{+2} ions respectively.

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94. What is standard hydrogen electrode?

A: 1) Standard hydrogen electrode is a reference electrode

2) H_2 gas adsorbed on platinum at one atmosphere placed in contact with 1MHCl concentration in solution is treated as standard hydrogen electrode (SHE).

3) Potential of this electrode is taken as zero.

4) By using the standard hydrogen electrode (SHE) we can measure the potential of single electrode.

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95. Write the Nernst equation for the EMF of the cell Ni(s) / Ni^{2+}(aq) // Ag^{+}(aq) / Ag(s)

A: 1) $E_{cell} = E^0_{cell} - \frac{RT}{2F} ln \frac{[Ni^{2+}]}{[Ag^{+}]^2}$ (or) $E_{cell} = E^0_{cell} - \frac{0.059}{2} log \frac{[Ni^{2+}]}{[Ag^{+}]^2}$

2) **Anode:** $Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^{-}$.....(1)

3) **Cathode:** $2Ag^{+}(aq) + 2e^{-} \longrightarrow 2Ag(s)$ .....(2);

4) **Net Equation:** $Ni(s) + 2Ag^{+}(aq) \longrightarrow Ni^{2+}(aq) + 2Ag(s)$

3 CHEMICAL KINETICS

2 Marks

96. What is a pseudo first order reaction? Give one example. **[BMP]** [AP 16,23]

A: 1) **Pseudo first order reaction** is a reaction in which one of the reactants, is present in large amount and its concentration does not get altered during the course of the reaction and it behaves as first order reaction.

2) **Ex: Hydrolysis of Ethyl acetate:** $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$

Rate of reaction $= k[CH_3COOC_2H_5]$

Here the H_2O is taken in excess. So, the rate doesn't depend on $[H_2O]$

---

97. Give two examples for gaseous first order reactions.

A: 1) $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$, order =1 [Decomposition of N_2O_5]

2) $C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$, order =1 [Hydrogenation of ethene]

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98. Give two examples for zero Order reactions. [AP 19]

A: 1) $H_2 + Cl_2 \xrightarrow{h\gamma} HCl$, rate= $k[H_2]^0[Cl_2]^0$, order =0

2) $2HI \xrightarrow{gold} H_2 + I_2$, rate = $k[HI]^0$, order =0

---

99. Define molecularity of a reaction with an example.

A: 1) **Molecularity of a reaction:** It is the number of reacting molecules taking part in a single elementary step.

2) **Examples:**

$C_2H_5Cl \rightarrow C_2H_4 + HCl$ (Unimolecular reaction)

$H_2 + I_2 \rightarrow 2HI$ (Bimolecular reaction)

$2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ (Trimolecular reaction)

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100. Write the equation for the rate of the reaction $5Br^-_{(aq)} + BrO^-_{3(aq)} + 6H^+_{(aq)} \rightarrow 3Br_{2(aq)} + 3H_2O(l)$

A: Rate $= -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t} = +\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = +\frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$

---

101. What is rate law? Illustrate with an example.

A: 1) **Rate Law:** It is an experimentally determined expression which relates the rate of reaction with concentration of reactants.

2) **Consider a hypothetical reaction:** $aA + bB \rightarrow$ products,

The rate law expression for this reaction is, Rate $= k[A]^m[B]^n$.

The order of the above reaction is $(m+n)$.

102. What are elementary reactions?

A: 1) The reactions taking place in one step are called elementary reactions.

2) **Ex:** $2HI \rightarrow H_2 + I_2$

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103. What are complex reactions? Name one complex reaction.

A: 1) **Complex reactions:** These are the reactions which involve sequence of elementary reactions for the conversion of reactants into products.

2) **Ex 1:** Oxidation of ethane to CO_2 and H_2O passes through a series of intermediate steps. Here alcohol, aldehyde and acid are formed.

Ex 2: $2H_2O_2 \rightarrow 2H_2O + O_2$

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104. What is half-life of a reaction? Illustrate your answer with an example.

A: 1) **Half-life of a reaction ($t_{1/2}$):** It is the time required to reduce half of the initial concentration of reactant.

2) For a zero order reaction, rate constant is given by $k = \frac{[R_0]-[R]}{t}$

At $t = t_{1/2}$ we have $[R] = \frac{[R_0]}{2}$

$\therefore$ The rate constant at $t_{1/2}$ is $k = \frac{[R_0]-\left[\frac{R_0}{2}\right]}{t_{1/2}} \Rightarrow k = \frac{[R_0]}{2[t_{1/2}]} \Rightarrow t_{1/2} = \frac{[R_0]}{2k}$

3) For a first order reaction, $t_{1/2} = \frac{0.693}{k}$

---

105. Calculate the overall order of a reaction which has the rate expression

(a) Rate = $k[A]^{1/2}[B]^{3/2}$ **(b) Rate = $k[A]^{3/2}[B]^{-1}$**

Sol: (a) Rate = $k[A]^x[B]^y$; Order = $x + y$ $\therefore$ Order = $\frac{1}{2}+\frac{3}{2}=\frac{4}{2}=2$ i.e., second order

(b) Order = $\frac{3}{2}+(-1)=\frac{3}{2}-1=\frac{1}{2}$ i.e., half order.

---

106. A first order reaction is found to have a rate constant $k = 5.5 \times 10^{-14}\ s^{-1}$. Find the half-life of the reaction. [AP 15]

Sol: 1) Half-life for a first order reaction is $t_{1/2} = \frac{0.693}{k}$

2) $\therefore t_{1/2} = \frac{0.693}{5.5\times10^{-14}s^{-1}} = 1.26\times10^{13}s$

4 The d&f-Block Elements

2+2 Marks

107. What are transition elements? Give examples.

A: 1) **Transition elements:** These are the elements with partially filled d-orbital either in their atoms or ions.

2) **Ex:** Chromium (Cr 24) : [Ar] $4s^1 3d^5$; for Cr^{+3} : [Ar]$4s^0 3d^3$.

Manganese (Mn 25) : [Ar]$4s^2 3d^5$. for Mn^{+2} : [Ar]$4s^0 3d^5$.

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108. Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic? [BMP] |AP 23|

A: 1) Electronic configuration of $Zn^{+2}$is[Ar] $4s^0 3d^{10}$

It has no unpaired electrons. So it is diamagnetic.

2) E.C of Mn^{+2} is [Ar]$4s^0 3d^5$

It has five unpaired electrons. So it is paramagnetic.

Atomic numbers
Zn = 30;Mn = 25;Ar = 18

☹ Zn-Paired-Dia
☺ Mn-Unpaired-Para

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109. Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion. [BMP] |AP 17,22,24|

A: 1) Electronic configuration of Fe^{+2} ion = [Ar] $3d^6 4s^0$

↿⇂	1	1	1	1

The number of unpaired electrons is n=4.

2) ∴ 'Spin only' magnetic moment $(\mu) = \sqrt{n(n+2)} = \sqrt{4(4+2)} = \sqrt{24} = 4.9$ BM

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110. Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why? |AP 16,25|

A: 1) E.C of Cu^{+2} is[Ar] $4s^0 3d^9$. It contains one unpaired electron.

Hence it exhibits blue colour in aqueous solution.

2) E.C of Zn^{+2} is [Ar] $4s^0 3d^{10}$.

It contains no unpaired electron. Hence it is colourless in aqueous solution.

Atomic numbers
Cu =29
Zn = 30

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111. What is an alloy? Give example. |AP 18|

A: 1) **Alloy:** It is a homogeneous mixture of a metal with other metals or metalloids or non-metals, having 'similar physical properties' to that of the original metal.

Alloys add more desirable properties to the original metal.

2)**Ex 1:** Gold is an alloy in Jewellery; Steel is an alloy of Iron and Carbon

Ex 2: Brass: 60-80% Cu; 20-40% Zn; Bronze: 75-90% Cu; 10-25% Sn

112. What is mischmetal? Give its composition and uses. [AP 16,19]

A: **1) Mischmetal** is a well known alloy of Lanthanoid metal.

2) Composition: 95% Lanthanoid metals (Ce, La, Nd, Pt), 5% iron and traces of S, C, Ca, Al.

3) Uses: Ignitium sources - Cigarette lighter, fire starter, bullets.

Mischmetal is a Bullet Metal

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113. What is actinoid contraction?

A: **1) Actinoid contraction:** It is the gradual decrease in the size of atoms or ions from left right in actinoid series.

2) It is due to poor screening effect of 5f orbitals.

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114. Why do the transition metals form a large number of complex compounds?

A: Transition metals form a large number of complex compounds due to their **small size, higher effective nuclear charge** and presence of **partially filled d-orbitals.**

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115. How do transition metals exhibit catalytic activity?

A: 1) Transition elements exhibit catalytic activity due to

(i) Variable oxidation states

(ii) Free valencies on their surfaces (iii) presence of **partially filled d-orbitals.**

2) Ex: Fe is the catalyst used in Haber's process in the manufacture of ammonia.

V_2O_5 is the catalyst used in Contact process in the manufacture of H_2SO_4.

Pt is the catalyst used in Ostwald's process in the manufacture of nitric acid.

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116. Why is it difficult to obtain M^{3+} oxidation state in Ni,Cu and Zn

A: 1) The element Ni, Cu and Zn are stable in +2 oxidation state and hence difficult to obtain +3 oxidation state.

2) Reason for +2 oxidation stability:

Ni and Cu in +2 oxidation state (Ni^{+2}) are stable due to highest negative enthalpy of hydration.

Zn in +2 oxidation state (Zn^{+2}) is stable due to completely filled d-subshell.

5 COORDINATION COMPOUNDS

2 Marks

117. What are coordination compounds? Give two examples.

A: **1) Coordination compounds** are the metal atoms that are bond to few anions or neutral molecules by sharing electrons through coordinate covalent bonds.

2) Ex: $[Co(NH_3)_6]Cl_3$, $K_4[Fe(CN)_6]$

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118. What is a double salt? Give example.

A: **1) Double salts** contain two or more compounds which are stable in solid state. But they dissociate into its constituent ions when dissolved in water . In this process, the individual properties of constituent ions are not lost.

2) Ex: Mohr salt - $FeSO_4$, $(NH_4)_2SO_4$. $6H_2O$

Potash alum - $K_2SO_4.Al_2(SO_4)_3.24H_2O$

Carnallite - $KCl.MgCl_2.6H_2O$

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119. What is the difference between a double salt and a complex compound?

A: 1) A double salt differs from complex compound in its dissociation.

2) A double salt dissociates completely into its constituent ions when dissolved in water whereas a complex compound do not dissociate into its constituent ions when dissolved in water.

3) **Ex: Double salt:** Carnalite $KCl.MgCl_2.6H_2O \rightarrow K^+ + Mg^{+2} + 2Cl^-$

Complex salt: Potassium ferrocyanide $K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$

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120. What is a chelate ligand ? Give example.

A: **1) Chelate ligand:** It is a ring-like structure obtained when a bidentate or a polydentate ligand uses its two or more donor atoms to bind a single metal ion.

2) Ex: Oxalate $(C_2O_4^{-2})$, en-(ethylene diammine) $\begin{pmatrix} CH_2 - NH_2 \\ | \\ CH_2 - NH_2 \end{pmatrix}$

---

121. What is an ambidentate ligand? Give example. **[BMP]** [AP 16]

A: **1) Ambidentate ligand:** It is a ligand which contains two donor atoms but only one of them forms a coordinate bond at a time with central metal atom / ion.

2) Ex:

$M \leftarrow SCN$	$M \leftarrow NCS$
thiocyanato	isothiocyanato
(S donor atom)	(N donor atom)

122. $CuSO_4.5H_2O$ is blue in colour where as anhydrous $CuSO_4$ is colourless. Why?

A: 1) In $CuSO_4.5H_2O$, water acts as ligand. As a result it causes crystal field splitting. Hence, d-d transition is possible in $CuSO_4.5H_2O$ and it shows colour.

2) In the anhydrous $CuSO_4$ due to the absence of water (ligand), crystal field splitting is not possible. Hence it is colourless.

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123. What is the oxidation state of cobalt in (i) $K[Co(CO)_4]$ and (ii) $[Co(NH_3)_6]^{3+}$

A: i) $K[Co(CO)_4]$: Oxidation state of cobalt is –1

ii) $[Co(NH_3)_6]^{3+}$: Oxidation state of cobalt is +3

---

124. $[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic. Why?

A: 1) $[Cr(NH_3)_6]^{3+}$ is **paramagnetic** due to the **presence** of three **unpaired electrons.**

2) $[Ni(CN)_4]^{2-}$ is **diamagnetic** due to the **absence** of **unpaired electrons.**

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125. $[Fe(CN)_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$ are of different colours in dilute solutions. Why?

A: 1) In both the complexes, Fe has +2 oxidation state . It has four unpaired electrons.

In the presence of **weak H_2O ligand**, the unpaired electrons **do not pair up.**

In the presence of **strong CN^- ligand**, the unpaired electrons **pair up .**

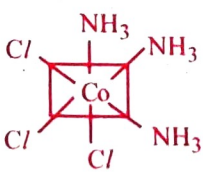
2) They have **different colours** due to the **difference in the number of unpaired electrons.**

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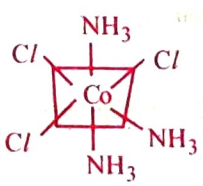
126. How many geometrical isomers are possible in the following coordination entities? (i) $[Cr(C_2O_4)_3]^{3-}$ (ii) $[CO(NH_3)_3Cl_3]$

A: i) No geometrical isomers are possible for $[Cr(C_2O_4)_3]^{-3}$.

ii) Two geometrical isomers are possible for $[CO(NH_3)_3Cl_3]$



Facial (fac)



Meridonial (mer)

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6 HALOALKANES & HALOARENES

2 Marks

127. What are Enantiomers? **[BMP][AP 16,23]**

A: 1) **Enantiomers** are a pair of stereo isomers which are non -super impossible mirror images.

2) **Ex:** d-Lactic acid & *l*-Lactic acid

☺ బింబం-ప్రతిబింబం ☺

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128. What type of isomerism is exhibited by o-, m- and p-dichlorobenzenes?

A: Isomerism exhibited by o-, m- and p-dichlorobenzenes is **Position isomerism.**

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129. What are ambident nucleophiles? **[AP 16,22,23]**

A: 1) **Ambident nucleophiles: These** are the nucleophiles with two donor atoms.

2) **Ex:** Cyanide ion, Nitrite ion.

Ambi means
☺ **Both** ☺

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130. What is the stereochemical outcome of S_N1 and S_N2 reactions?

A: 1) The stereochemical outcome of S_N1 reaction is a **racemic mixture** of two optical isomers (50:50).

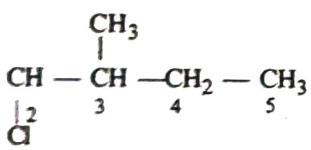
2) The stereochemical outcome of S_N2 reaction is **Walden inversion** (inversion of configuration).

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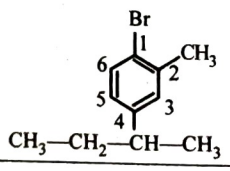
131. Write the structure of the following compounds **[AP 15]**

(i) 2-chloro-3-methylpentane **(ii)1-bromo-4-sec-butyl-2-methylbenzene**

A: (i)



(ii)



---

132. Write the isomers of the compound having molecule formula C_4H_9Br?

A: 1) $CH_3—CH_2—CH_2—CH_2Br$ 1–Bromobutane

2) $CH_3—CH_2—\underset{\displaystyle Br}{\underset{|}{CH}}—CH_3$ 2–bromobutane

3) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2Br$ 1–bromo-2 methyl propane

4) $CH_3—\underset{\displaystyle Br}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}—CH_3$ 2–bromo-2 methyl propane

133. Which compound in each of the following pairs will react faster in S_N2 reaction with OH?
(i) CH_3Br or CH_3I (ii) $(CH_3)_3CCl$ (or) CH_3Cl

A: i) CH_3I reacts faster because C-I has **less dissociation energy** than that of C- Br bond.
ii) CH_3Cl reacts faster because it has **less steric hinderance** as compared to $(CH_3)_3CCl$.

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134. Between $C_6H_5CH_2Cl$ and $C_6H_5CHClC_6H_5$, which is more easily hydrolysed by aqueous KOH?

A: 1) $C_6H_5CHClC_6H_5$ will be more easily hydrolysed than $C_6H_5CH_2Cl$.
2) Because its carbocation(2°) is stabilised by two phenyl groups (C_6H_5-) due to resonance under S_N1 reaction.

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135. Explain why the alkyl halides though polar are immiscible with water?

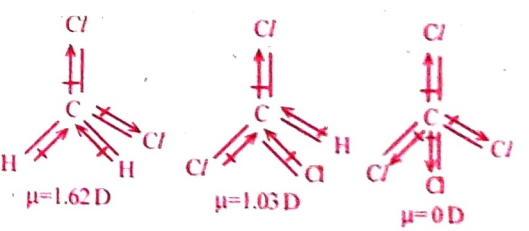
A: Alkyl halides (R-X) are polar due to the presence of polar C-X bond.
Alkyl halides are immiscible with water because
a) Alkyl halides are not able to form hydrogen bonds with water molecules.
b) Alkyl halides are not able to break hydrogen bonds in water.

---

136. Which one of the following has highest dipole moment?
(i) CH_2Cl_2 (ii) $CHCl_3$ (iii) CCl_4

A: CH_2Cl_2 has highest dipole moment

i) In CH_2Cl_2, the resultant dipolements of two C-Cl bonds is reinforced by two C-H bond moments.
Its dipole moment is $\mu = 1.62D$.



ii) In $CHCl_3$, the resultant dipole moment of two C-Cl bonds is opposed by resultant dipolemoment of C-Cl and C-H bonds. Here the net dipolemoment is $\mu = 1.03D$

iii) CCl_4 is a symmetrical molecule . Its dipole moment is $\mu = 0$.



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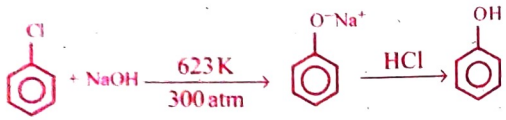
7 ALCOHOLS, PHENOLS & ETHERS

2 Marks

137. Give the reagents used for the preparation of phenol from chlorobenzene. [BMP]

A: 1) Phenol can be obtained by heating chlorobenzene in the presence of a catalyst, with 10% NaOH solution and HCl at 350°C under 200-300 atm pressure.

2) Reaction:



---

138. Explain why propanol has higher boiling point than that of the hydrocarbon butane.

A: 1) The boiling point of propanol is higher because of the presence of intermolecular hydrogen bonding in the molecules. But, it is not present in butane due to the absence of polar –OH group.

2) The **boiling point (391K) of propanol** is **more** than that of **butane (309K).**

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139. Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact.

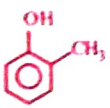
A: 1) Alcohols can form H-bonds with water and break the H-bonds already existing between water molecules. Hence, they are soluble in water.

2) But hydrocarbons cannot form H-bonds with water and hence they are insoluble in water.

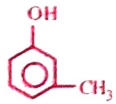
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140. Give the structures and IUPAC names of monohydric phenols of molecular formula C_7H_8O.

A:



2-methylphenol
(o-Cresol)



3-methylphenol
(m-cresol)



4-methylphenol
(p-cresol)

---

141. Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason.

A: Ethers cannot be prepared by the dehydration of secondary (or) tertiary alcohols, because alkenes are formed easily in these reactions. This happens due to **steric hinderance.**

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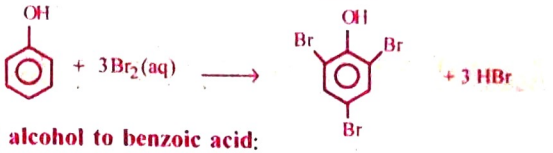
142. Write the equations for the following reactions

i) Bromination of phenol to 2,4,6-tribromophenol

ii) Benzyl alcohol to benzoic acid

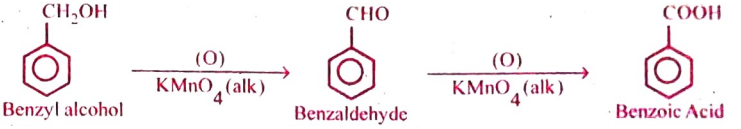
A: **i) Bromination of phenol to 2,4,6-tribromophenol:**

Phenol when treated with excess of aqueous bromine solution forms 2,4,6-tribromophenol.



ii) Benzyl alcohol to benzoic acid:

Benzyl alcohol oxidises into Benzaldehyde and then to Benzoic acid in the presence of alkaline $KMnO_4$.



---

143. Write the structures for the following compounds
1) Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane

A: 1) **Ethoxy Ethane :** $C_2H_5 - O - C_2H_5$
2) **Ethoxybutane :** $C_2H_5 - O - CH_2 - CH_2 - CH_2 - CH_3$
3) **Phenoxy ethane or Ethoxy benzene or Ethyl phenyl ether :** $C_6H_5 - O - C_2H_5$

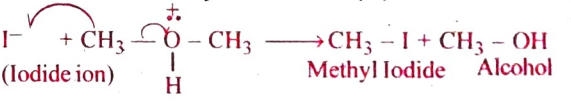
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144. Write the mechanism of the reactions of HI with methoxymethane.

A: 1)The ether molecule is initially protonated by the halogen acid (HI) to form protonated ether.



2)The protonated ether is then attacked by the halide ion (I^-) which acts as nucleophile as follows:



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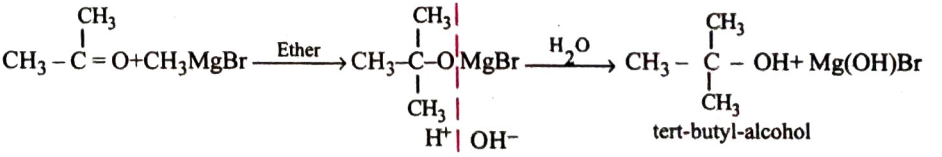
145. Name the reagents used in the following reactions:
i) Oxidation of primary alcohol to carboxylic acid
ii) Oxidation of primary alcohol to aldehyde

A: i) Acidified $KMnO_4$ (or) acidified $K_2Cr_2O_7$ is used in the oxidation of primary alcohol to carboxylic acid
ii) CrO_3 (or) pyridine chlorochromate (pcc) is used as a reagent in the oxidation of primary alcohol to aldehyde.

---

146. Identify the reactant needed to form t-butyl alcohol from acetone.

A: **CH_3MgBr** (Methyl magnesium bromide) is needed to form t-butyl alcohol from acetone.



8 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

2 Marks

147. List the reagents suitable to reduce carboxylic acid to alcohol. [BMP]

A: **LiA/H_4 (or) copper chromate** are the needed reducing reagents .

$$CH_3COOH + 4(H) \xrightarrow{LiAlH_4} C_2H_5OH + H_2O$$

---

148. Write the reaction showing α-halogenation of carboxylic acid and give its name.

A: **HVZ Reaction:**

Carboxylic Acid having atleast one α- hydrogen is treated with Halogens (Br (or) C*l*) in the presence of red phosphorus undergo Halogenation to form α-Halo Carboxylic Acid.

$$\underset{\text{Carboxylic acid}}{R-CH_2-COOH} \xrightarrow[(ii)\ H_2O]{(i)\ X_2/\text{Red phosphorus}} \underset{\alpha\text{–halo carboxylic acid}}{R-\overset{X}{\overset{|}{C}}H-COOH} + HX$$

---

149. Arrange the following compounds in increasing order of their property indicated.

i) acetaldehyde, acetone and t-butylmethyl ketone reactivity towards HCN

ii) fluoroacetic acid, monochloroacetic acid, acetic acid and dichloroacetic acid (acid strength)

A: i) **Reactivity towards HCN:** t-butylmethyl ketone < acetone < acetaldehyde.

ii) **Acid strength:** dichloroacetic acid >fluoroaceticacid > monochloroacetic acid > acetic acid.

---

150. How do you distinguish acetophenone and benzophenone?

A: **1) Acetophenone** undergoes iodoform reaction due to the **presence of acetyl group CH_3–CO .**

2) But **benzophenone** does not undergo iodoform reaction due to the **absence** of **CH_3–CO** group

$$\underset{\text{Acetophenone}}{C_6H_5COCH_3} + 3I_2 + 4NaOH \xrightarrow{\text{heat}} C_6H_5COONa + \underset{\text{Iodoform}}{CHI_3} + 3H_2O + 3NaI$$

---

151. An organic acid with molecular formula $C_8H_8O_2$ on decarboxylation forms Toluene. Identify the organic acid.

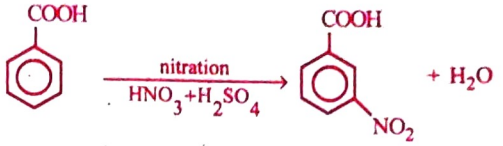
A: The organic acid formed is phenyl acetic acid (2-Phenyl ethanoic acid)



152. Explain the position of electrophilic substitution in benzoic acid.

A: 1) COOH group on benzene, deactivates the ring at ortho and para positions. Hence substitution takes place at meta position.

2)Ex:



(m-nitrobenzoic acid)

---

153. Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol. [AP 22][IPE'14]

A: **1) The relative values of k_a and pk_a indicate the acidic strength:**

Higher k_a value and lower pk_a value indicates that the acid is strong.

2) Order of acidic strength: Chloroacetic acid > benzoic acid > acetic acid > phenol

The acidic strength of acetic acid, chloroacetic acid, benzoic acid and phenol:

Acid	k_a (298 k)	pk_a (298 k)
1) CH_3COOH (Acetic acid)	1.75×10^{-5}	4.76
2) $Cl–CH_2–COOH$ (Chloroacetic acid)	1.36×10^{-3}	2.87
3) C_6H_5COOH (Benzoic acid)	6.3×10^{-5}	4.0
4) C_6H_5OH (Phenol)	1.1×10^{-10}	10

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154. Write equations showing the conversion of

i) acetic acid to acetyl chloride **ii) benzoic acid to benzamide**

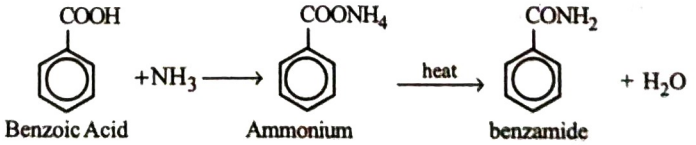
A: **i) acetic acid to acetyl chloride:**

$$3CH_3COOH + PCl_3 \longrightarrow 3CH_3COCl + H_3PO_3$$

$$CH_3COOH + PCl_5 \longrightarrow CH_3COCl + POCl_3 + HCl$$

$$CH_3COOH + SOCl_2 \longrightarrow CH_3COCl + SO_2 + HCl$$

ii) benzoic acid to benzamide



Benzoic Acid — Ammonium benzoate — benzamide

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9 AMINES

2 Marks

155. Arrange the following bases in increasing order of their basic strength. Aniline, p-nitroaniline and p-toluidine. [BMP]

A: Increasing order (low to high) of basic strength: p-nitroaniline < Aniline < p-toluidine.

---

156. Write equations for carbylamine reaction of an aliphatic amine.

A: 1) When a primary amine is heated with alcoholic caustic potash and chloroform, an offensive smelling compound called carbylamine (Ethylisocyanide) is formed. This test is used for the detection of primary amines.

2) Reaction: $\underset{\text{Ethanamine}}{C_2H_5NH_2} + CHCl_3 + 3KOH \xrightarrow[\Delta]{\text{alc.}} \underset{\text{Ethyl carbylamine}}{C_2H_5NC} + 3KCl + 3H_2O$

---

157. Gabriel phthalimide synthesis exclusively forms primary amines . Explain.

A: **Gabriel phthalimide** synthesis is preferred for synthesizing aliphatic primary amines because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.

---

158. Arrange the following bases in decreasing order of pK_b values
$C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$.

A: Decreasing order of pK_b : $C_6H_5NH_2 > C_6H_5NHCH_3 > C_2H_5NH_2 > (C_6H_5)_2NH$

pK_b values: 9.38 9.30 3.29 3.00

[Decreasing order of Basicity: $(C_6H_5)_2NH > C_2H_5NH_2 > C_6H_5NHCH_3 > C_6H_5NH_2$]

---

159. Write the IUPAC names of the following compounds and classify into primary, secondary and tertiary amines.

(i) $(CH_3)_2CHNH_2$ (ii) $CH_3(CH_2)_2NH_2$ (iii) $(CH_3CH_2)_2NCH_3$.

A: (i) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-NH_2$ Propan-2-amine (Primary amine)

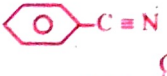
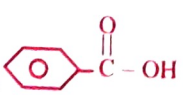
(ii) $CH_3-CH_2-CH_2-NH_2$ 1-Propanamine (Primary amine)

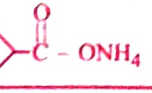
(iii) $CH_3-CH_2-\underset{\displaystyle CH_2-CH_3}{\underset{|}{N}}-CH_3$ N-ethyl-N-Methyl-ethanamine (Tertiary amine)

---

160. Give structures of A,B and C in the following reaction.

$C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow{NH_3} C$

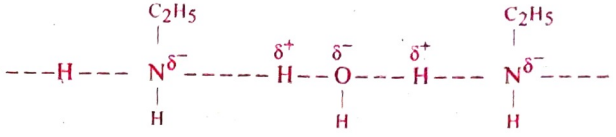
A: **A** : C_6H_5CN (or)  **B** : C_6H_5COOH(or) 

C : $C_6H_5COONH_4$ (or) 

161. Explain why ethylamine is more soluble in water where as aniline is sparingly soluble.

A: 1) Ethylamine forms hydrogen bonds with water molecules and hence dissolves in water.

2) Aniline, due to its large hydrocarbon part, develops hydrophobic character. As a result it does not dissolve in water.

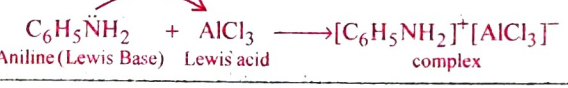


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162. Why aniline does not undergo Friedel-Craft reactions?

A: **1) Aniline** doesnot undergo Friedel-Crafts reaction. Actually, aniline being a Lewis base forms a complex with $AlCl_3$ which is a Lewis acid.

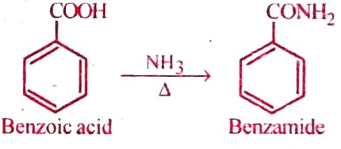
2) The amino group is not in a position to activate the benzene ring towards electrophilic substitution. Hence, alkylation (or) acylation which lead to Fridel Crafts reaction is not possible.



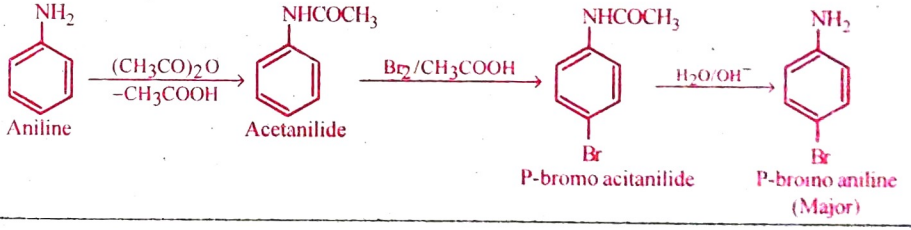
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163. Accomplish the following conversions:
(i) Benzoic acid to benzamide (ii) Aniline to p-bromoaniline

A: i) **Benzoic acid to Benzamide:**



ii) **Aniline to p-bromo aniline**:

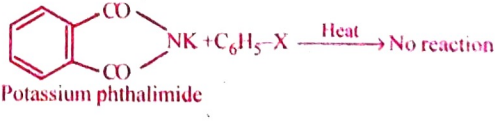


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164. Why aromatic primary amines cannot be prepared by Gabriel phthalimide?

A: 1) Aryl halides does not react with potassium phthalimide. Because C-X bond in haloarene is difficult to be cleaved due to partial double bond character.

2) They do not undergo nucleophilic substitution with the anion formed by phthalimide. So, aromatic primary amines cannot be prepared by Gabriel phthalimide synthesis.



10 BIOMOLECULES

2+2Marks

165. What are essential and non-essential amino acids? Give one example for each. [BMP]

A: 1) **Essential Amino acids:** These are the amino acids which cannot be synthesised in the body and must be taken in diet. **Ex:** Valine, leucine, phenylalanine .

2) **Non-essential amino acids:** These are the amino acids which can be synthesised in the body **Ex:** Glycine alanine, glutamic acid.

---

166. Name the vitamin responsible for the coagulation of blood.

A: **Vitamin K** is responsible for the coagulation of blood.

---

167. Differentiate between globular and fibrous proteins. [BMP][AP 20]

A:

Globular Proteins	Fibrous Proteins
1) Polypeptide chains are arranged as coils.	1) Polypeptide chains are parallel to each other.
2) They have spherical shape.	2) They have thread like structure.
3) They are water soluble.	3) They are insoluble in water.
4) **Ex:** Insulin, albumins etc.,	4) **Ex:** Keratin, Myosin etc.,

---

168. What is glycogen? How is it different from starch?

A: 1) **Glycogen** is a polysaccharide which is stored in animal body (liver, muscles, brain).

Starch is a polysaccharide which is stored in plants.

2) **Glycogen** is more highly branched than Starch (amylopectin).

---

169. What are anomers?

A: 1) **Anomers** are those isomers which differ in orientation of –OH at C-1 (first carbon)'
2) **Ex:** α-D-glucose and β-D-glucose

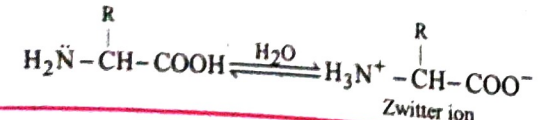
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170. What is Zwitter ion? Give an example. [AP 15,22]

A: 1) **Zwitter ion:** It is a 'dipolar neutral ion' containing negative and positive charges.

2) **Ex:** Free amino acids in cells.

3) In Amino acids, proton transfer takes place from –COOH group to NH_2 group. This results in the formation of Zwitter ion.



☺ U C ☺ Zwitter has Twitter

171. Write the names of three types of RNA.

A: **Three types of RNA:**

a) m-RNA (messenger RNA) b)r-RNA (ribosomal RNA) c) t-RNA (transfer RNA)

---

172. Why are sugars classified as reducing and non-reducing sugars?

A: 1) Sugars are classified as reducing and non-reducing sugars to **test** whether the sugar has a free aldehyde or ketone group that can act as a reducing agent.

2) **Reducing sugars** reduce Fehling's solution or Tollen's reagent .
They contain free aldehyde or Ketonic group. **Ex:** Glucose & Maltose

3) **Non-reducing sugars** do not reduce Fehling's solution or Tollen's reagent.
They do not ~~have~~ free aldehyde or ketonic group. **Ex:** Sucrose & Cellulose

---

173. Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each.

A: **Classification of Carbohydrates (Saccharides) on the basis of their hydrolysis:**

a) **Monosaccharides** cannot be broken further into smaller units on hydrolysis
Ex: Glucose and fructose.

b) **Disaccharides** give 2 units of monosaccharides on hydrolysis.
Ex: Sucrose, lactose

c) **Oligosaccharides**: These carbohydrates give 2-10 units of monosaccharides on hydrolysis.
Ex: Raffinose(which gives glucose, fructose and galactose on hydrolysis)

d) **Polysaccharides**: These carbohydrates give large number of monosaccharides on hydrolysis.
Ex: Starch, cellulose etc.,

---

174. What is the difference between a nucleoside and a nucleotide?

A: These two are fundamental building blocks of nucleic acids DNA/ RNA

Nucleoside	Nucleotide
1) Nucleoside = Base + Sugar	1) Nucleotide = Base + Sugar + Phosphoric acid
2) Nucleosides are primarly structural components.	2) Nucleotides also act as energy units (ATP)
3) Nucleoside is a unit formed by attachment of a base to C-1 position of sugar.	3) Nucleotide is linked to phosphoric acid to C-5 position of sugar.
4) **Ex:** Adenine + Ribose= adenosine	4) **Ex:** Adenosine + Phosphoric acid = Adenylic acid